The logarithmic coefficients in Federbush's Awesome Conjecture

MathDB 349137

Principia Math In Progress

September 18, 2026

Abstract

We prove both coefficient assertions in Conjecture B.1 of Federbush's arXiv:2012.10927v8 for every order. A coefficient recurrence determines the highest-degree terms of the unperturbed logarithm. Finite shifts change only lower-degree terms. All arguments take place in rings of formal power series; no asymptotic interchange is required.

Task 1. Full solution

Statement and the source normalization

Write $\varepsilon = n^{-1}$ and $(j)_{\underline{z}} = j(j-1)\cdots(j-z+1)$. In the notation of equations (3.1)–(3.5) of [1], let

$$T_r(y) = \frac{2(r-1)}{r-2+r\sqrt{1-4(r-1)y}} = 1 + ry + \sum_{s\geq 2} u_s(r) y^s, \tag{1}$$

$$f(x) = rx - \sum_{s\geq 2} \frac{u_s(r)}{s}(-x)^s = \sum_{s\geq 1} f_s x^s, \tag{2}$$

$$M_j(n) = [x^j] e^{nf(x)}, \qquad A_j(\varepsilon) = \frac{j!}{(nr)^j} M_j(n) = \sum_{h\geq 0} a_h(r,j)\varepsilon^h. \tag{3}$$

The square root has constant term 1. These identities may first be read over $\mathbb{Q}(r)$. At $r = 1$, use the equivalent rationalized expression

$$T_r(y) = \frac{r\sqrt{1-4(r-1)y} - r + 2}{2(1-r^2 y)},$$

which gives $T_1(y) = (1-y)^{-1}$. Every specialization to a positive integer r is therefore defined. The coefficients $a_h(r,j)$ have their natural polynomial extensions in j, made explicit below.

Theorem 1. *Fix constants c_i and integers $z_i \geq 2$, and put*

$$F_j = A_j + \sum_i \frac{c_i}{r^{z_i}} \varepsilon^{z_i} (j)_{\underline{z_i}} A_{j-z_i}.$$

For every $h \geq 1$, the coefficient $[\varepsilon^h] \log F_j$ has degree at most $h+1$ in j, and

$$[j^{h+1}\varepsilon^h] \log F_j = \frac{r^{-h} - 2}{h(h+1)}. \tag{4}$$

Thus both assertions (B.2) and (B.3) in [1] hold without the finite-order restrictions in that paper's Theorem B.1.

The sum over i can be finite or coefficientwise defined, for example with finitely many indices at each bounded shift. Constants are independent of j and n. This is the formal coefficient convention of the source.

A known degree bound, with a proof

The following bound is already Theorem 3.1 of [1]; its proof there uses Robin Chapman's Stirling-number identity. We include a formal connected-graph proof to make the present argument self-contained.

Lemma 2. *For any $f(x) = rx + O(x^2)$ over a characteristic-zero field with $r \neq 0$, the coefficients in (3) are polynomials in j of degree at most $2h$, and*

$$L_j := \log A_j = \sum_{h\geq 1} \ell_h(j)\varepsilon^h, \qquad \deg \ell_h \leq h+1.$$

Proof. For each subset B of $[j]$ of size $s \geq 2$, assign weight $w(B) = s! f_s r^{-s} \varepsilon^{s-1}$. Expanding the exponential in (3) shows that A_j is the sum of the products of these weights over all families of pairwise disjoint such subsets, including the empty family. More explicitly,

$$a_h(r,j) = \sum_{\substack{m_2,m_3,\ldots\geq 0 \\ \sum (s-1)m_s = h}} (j)_{\underline{\sum s m_s}} \prod_{s\geq 2} \frac{(f_s/r^s)^{m_s}}{m_s!}. \tag{5}$$

Here $\sum s m_s \leq 2h$, proving the first bound and defining the polynomial extension.

For completeness, the connected-graph expansion of its logarithm is

$$\log A_j = \sum_{m\geq 1} \frac{1}{m!} \sum_{B_1,\ldots,B_m} \prod_{i=1}^{m} w(B_i) \sum_{H \text{ connected graph on } [m]} \prod_{\{i,k\}\in E(H)} \left(-\mathbf{1}_{B_i\cap B_k\neq\varnothing}\right). \tag{6}$$

The subsets in the tuple can repeat. To verify this identity, expand $\prod_{i<k}(1 - \mathbf{1}_{B_i\cap B_k\neq\varnothing})$ in the ordered-tuple expression for A_j. Its expansion sums over all graphs on the tuple. Decomposing each graph uniquely into connected components is precisely the formal exponential of the right side of (6). Each weight has positive ε-order, so all coefficient calculations are finite.

A nonzero connected term of order h has $h = \sum_i (|B_i| - 1)$ and a connected intersection graph. Order its subsets along a rooted spanning tree. Each new subset meets the previous union, so the total number v of labels in the union is at most

$$|B_1| + \sum_{i=2}^{m} (|B_i| - 1) = h + 1.$$

Grouping tuples by their union size gives a finite linear combination of $\binom{j}{v}$ with $v \leq h+1$. This proves the logarithmic bound. □

Determining the highest coefficients

Set

$$b_h = [j^{h+1}]\ell_h(j), \qquad B(t) = \sum_{h\geq 1} b_h t^{h+1}.$$

Differentiating $e^{nf(x)}$ with respect to x gives $jM_j = n\sum_{s\ge1} sf_sM_{j-s}$. After normalization and division by A_j, this becomes

$$j\varepsilon = \sum_{s\ge1} \frac{sf_s}{r^s}\varepsilon^s(j)_{\underline{s}} \exp(L_{j-s} - L_j). \tag{7}$$

For integer j, terms $s > j$ vanish. At any fixed ε-order, only finitely many s occur and both sides are polynomial in j; hence the identity also holds with j an indeterminate.

Substitute $j = t/\varepsilon$ in (7). Lemma 2 gives

$$\deg_j\big(\ell_h(j-s) - \ell_h(j)\big) \le h.$$

Consequently every $L_{j-s} - L_j$ after substitution lies in $\mathbb{Q}(r)[[t,\varepsilon]]$, with zero constant term, and its ε^0 coefficient is $-sB'(t)$. Also

$$\varepsilon^s(j)_{\underline{s}} = \prod_{a=0}^{s-1}(t - a\varepsilon).$$

This polynomial has total degree s, so the infinite sum remains coefficientwise defined in the two-variable power-series ring. Taking its ε^0 coefficient proves the exact formal identity

$$t = \sum_{s\ge1} sf_s(t/r)^s e^{-sB'(t)} = x(t)f'(x(t)), \qquad x(t) = \frac{t}{r}e^{-B'(t)}. \tag{8}$$

No limit involving large j or large n has been used.

From (1)–(2),

$$xf'(x) = 1 - T_r(-x), \qquad (1 - r^2y)T_r(y)^2 + (r-2)T_r(y) - (r-1) = 0.$$

The quadratic identity also holds at $r = 1$ by the rationalized formula. Substitute $y = -x(t)$ and $T_r(-x(t)) = 1 - t$ to obtain

$$r^2x(t)(1-t)^2 = t(r-t), \qquad x(t) = \frac{t(r-t)}{r^2(1-t)^2}.$$

Since $B'(0) = 0$, taking the formal logarithm in (8) gives

$$\begin{aligned} B'(t) &= \log\frac{t}{rx(t)} = 2\log(1-t) - \log(1-t/r) \\ &= \sum_{h\ge1} \frac{r^{-h} - 2}{h}\, t^h. \end{aligned}$$

Thus $b_h = (r^{-h} - 2)/(h(h+1))$, as required for A_j.

Why the shifts leave these terms unchanged

Let $\mathcal{R}$ consist of the formal series $\sum_{h\ge0} p_h(j)\varepsilon^h$ with $\deg p_h \le h$. It is a ring, and formal exponentiation of its elements with zero constant term, and logarithms of its elements with constant term 1, remain in $\mathcal{R}$.

For every fixed z, Lemma 2 gives $L_{j-z}-L_j \in \mathcal{R}$ with zero constant term. Hence $\exp(L_{j-z}-L_j) \in \mathcal{R}$. More explicitly, its ε^h coefficient has degree at most h. Multiplication by $\varepsilon^z(j)_{\underline{z}}$ preserves this bound relative to the new order. It follows that

$$\frac{F_j}{A_j} = 1 + \sum_i \frac{c_i}{r^{z_i}} \varepsilon^{z_i} (j)_{\underline{z_i}} \exp(L_{j-z_i} - L_j)$$

belongs to $\mathcal{R}$ and has constant term 1. Therefore

$$\deg_j[\varepsilon^h](\log F_j - \log A_j) \le h.$$

This proves Theorem 1. □

Verification, attribution, and scope

The source definitions, conjecture, and known general degree bound are credited to [1]. The highest-coefficient expression itself is Pernici's formula, discussed in [1] as equation (3.7). The present proof establishes it at all orders and proves its invariance under the stated perturbations.

The accompanying standard-library verifier checks the source quadratic, 156 exact coefficient normalizations, both unperturbed assertions through order 12 at four values of r, 192 perturbed coefficients, and a universal polynomial certificate for the inverse relation. These checks supplement the proof, rather than establish an unbounded assertion by computation. Review is by the authoring agent; no independent peer review or proof-assistant certification is claimed.

All 18 pages of the current primary were read. Its metadata, the author's mp_arc records, bounded public searches, and the checked claim indexes did not identify a journal final or a prior full claim for this target as of September 18, 2026. The 2022 extension [2] states a finite range, not an all-order theorem. This result concerns the formal coefficient conjecture; probabilistic and infinite-volume consequences require their own hypotheses.

Tasks 2–5. Proposed follow-up directions

Task 2. Derive explicit formulas and effective recurrences for the next diagonals $[j^{h-d}\varepsilon^h] \log F_j$, uniformly in h for fixed d.

Task 3. Revisit the proof of weak graph positivity in [1] and determine which probabilistic estimates suffice to remove its remaining parameter restrictions using Theorem 1.

Task 4. Analyze perturbation coefficients that scale with n, as in infinite-lattice applications, and determine the resulting change in the highest-degree generating function $B(t)$.

Task 5. Relate the shift proof to the coefficient-change formulation in Federbush's Genius Conjectures; determine whether it gives uniform bounds for the auxiliary corrections there.

References

[1] P. Federbush, *A PROOF of Weak Graph Positivity, for a Large Range of the Parameters*, arXiv:2012.10927v8, 2022. See Theorem 3.1, equations (3.1)–(3.7), and Conjecture B.1.

[2] P. Federbush, *Extension of Range of Validity of Pernici's* $1/n$ *Conjecture*, mp_arc 22-22, April 29, 2022.